Year 8 Mathematics

Unit 1: Number

1.3 Using Indices

Using Indices

STARTER · 5 MIN

Date: _____ **Title: Using Indices**

LEARNING OBJECTIVES

- To be able to use the laws of indices to simplify expressions with powers

STARTER *Answer in your book. No calculators.*

1. Work out 3^5
2. Work out 4^3
3. Work out 5^2
4. Work out 3^3

Starter — Answers

STARTER

Tick or correct in green pen

1. **243**

2. **64**

3. **25**

4. **27**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can apply the basic index laws for multiplying and dividing powers with the same base	☐
SILVER	I can simplify more complex expressions involving several steps, including brackets	☐
GOLD	I can use index laws to solve real-world problems	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Using Indices

INTRODUCTION · 3 MIN

When the base is the same, add powers to multiply and subtract powers to divide.

Multiply

Add the powers.

$a^m \times a^n = a^{m+n}$ $2^2 \times 2^3 = 2^5$

Divide

Subtract the powers.

$a^m \div a^n = a^{m-n}$ $2^5 \div 2^3 = 2^2$

Power of a power

Multiply the powers.

$(a^m)^n = a^{mn}$ $(2^2)^3 = 2^6$

Power 0 and 1

Anything to the power 0 is 1.

$a^0 = 1$ $a^1 = a$

Examples 1 and 2 — Multiplying and Dividing Powers

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint × : add the powers ÷ : subtract the powers

Example 1

Write as a single power of 2

$2^2 \times 2^3$

Example 2

Write as a single power of 2

$2^5 \div 2^3$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Same base: add the powers

$2^2 \times 2^3 = 2 \times 2 \times 2 \times 2 \times 2$

$= 2^{2+3}$

$\mathbf{= 2^5}$

Tip: count the 2s.

Example 2

Same base: subtract the powers

$2^5 \div 2^3 = 2^{5-3}$

$\mathbf{= 2^2}$

Tip: three of the 2s cancel.

Task 1

PRACTICE · 9 MIN

BRONZE *Write as a single power*

Hint × : add the powers ÷ : subtract the powers

1 $4^5 \times 4^2$

2 $7^3 \times 7$

3 $5^6 \div 5^2$

4 $3^8 \div 3^3$

5 $9^4 \times 9^4$

★ $6^7 \div 6$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1 4^{7}

2 7^{4}

3 5^{4}

4 3^{5}

5 9^{8}

★ 6^{6}

Examples 3 and 4 — Brackets and Several Steps

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint $(a^m)^n$: multiply the powers $a^0 = 1$

Example 3

Write as a single power of 2

$(2^2)^3$

Example 4

Write as a single power of 3

$$\frac{3^4 \times 3^5}{3^2}$$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Power of a power: multiply

$(2^2)^3 = 2^2 \times 2^2 \times 2^2$

$\mathbf{= 2^6}$

Tip: 2 × 3 = 6.

Example 4

Top first, then divide

$3^4 \times 3^5 = 3^9$

$\mathbf{3^9 \div 3^2 = 3^7}$

Tip: 4 + 5 − 2 = 7.

Task 2

PRACTICE · 9 MIN

SILVER *Write as a single power*

Hint $(a^m)^n$: multiply the powers $a^0 = 1$

1 $(4^3)^2$

2 $(5^2)^4$

3 $\dfrac{2^7 \times 2^3}{2^5}$

4 $\dfrac{5^6}{5^2 \times 5}$

5 7^0

★ $(3^2)^3 \div 3^4$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 4^6

2 5^8

3 2^5

4 5^3

5 1

★ 3^2

Examples 5 and 6 — Using Index Laws to Solve Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint Simplify, then work out Same base on both sides

Example 5

(2 marks)

Work out the value of

$$\frac{2^{17}}{2^{10}}$$

Example 6

(1 mark)

Find the value of n.

$$3^{n} \times 3^{4} = 3^{10}$$

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Simplify first, then work out

$$\frac{2^{17}}{2^{10}} = 2^{17-10}$$

$$= 2^{7}$$

$$\mathbf{= 128}$$

Tip: never work out 2^{17} first.

Example 6

Multiplying: the powers add

$$n + 4 = 10$$

$$\boldsymbol{n = 6}$$

Tip: check $3^{6} \times 3^{4} = 3^{10}$.

Task 3

PRACTICE · 9 MIN

GOLD *Exam-style: index laws*

Hint Simplify, then work out Same base on both sides

1 $\dfrac{5^9}{5^7}$

2 $\dfrac{10^8}{10^5}$

3 $2^n \times 2^3 = 2^9$. Find n.

4 $(7^n)^2 = 7^8$. Find n.

5 Write $4^3 \times 2^5$ as a single power of 2

★ How many 1 kB (2^{10} byte) files fit in 2^{16} bytes?

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1 $5^2 = 25$

2 $10^3 = 1\,000$

3 $n = 6$

4 $n = 4$

5 $(2^2)^3 \times 2^5$

$= 2^{11}$

★ $2^{16} \div 2^{10} \quad = 2^6 = 64$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Simplify $a^m \times a^n$ and $a^m \div a^n$

SILVER Write $(5^3)^4$ as a single power

GOLD Work out the value of $\frac{3^7}{3^5}$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE a^{m+n} and a^{m-n}

SILVER 5^{12}

GOLD $3^2 = 9$

Exam Practice — Index Laws

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint × : add the powers ÷ : subtract the powers

Question 1

(2 marks)

Write as a single power of 6.

(a) $6^3 \times 6^4$ (1)

.....

(b) $6^9 \div 6^4$ (1)

.....

Question 2

(2 marks)

Write as a single power.

(a) 8×8^5 (1)

.....

(b) 9^0 (1)

.....

Exam Practice — Several Steps

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint $(a^m)^n$: multiply the powers $a^0 = 1$

Question 3

(2 marks)

Write $(3^4)^2$ as a single power of 3.

.....

Question 4

(2 marks)

Simplify

$$\frac{7^5 \times 7^4}{7^3}$$

Give your answer as a power of 7.

.....

Exam Practice — Finding Powers

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint Simplify, then work out
Same base on both sides

Question 5

(2 marks)

Work out the value of

$$\frac{4^8}{4^6}$$

.....

Question 6

(3 marks)

Find the value of n in each.

(a) $5^n \times 5^3 = 5^{11}$ (1)

.....

(b) $(2^n)^3 = 2^{12}$ (1)

.....

(c) $8 \times 2^4 = 2^n$ (1)

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 6^7 — B1

(b) 6^5 — B1

Tip: add to ×, subtract to ÷.

Question 2

(a) 8^6 — B1

(b) $1 (= 9^0)$ — B1

Tip: $8 = 8^1$.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$4 \times 2 = 8$ **M1**

$\mathbf{3^8}$ **A1**

Tip: multiply the powers.

Question 4

$7^5 \times 7^4 = 7^9$ **M1**

$\mathbf{7^9 \div 7^3 = 7^6}$ **A1**

Tip: top first, then divide.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$4^{8-6} = 4^2$ — M1

$= 16$ — A1

Tip: simplify before you calculate.

Question 6

(a) $n = 8$ — B1

(b) $n = 4$ — B1

(c) $2^3 \times 2^4$: $n = 7$ — B1

Tip: $8 = 2^3$.